

1.1.

1. Reální část

Pozice $\|u+v\|^2 - \|u-v\|^2$ s použitím vzorce 8.32:

$$\|u+v\|^2 - \|u-v\|^2 = \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle}$$

$$\rightarrow (\|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle}) =$$

$$= 2\operatorname{Re}(\langle u, v \rangle) - (-2\operatorname{Re}(\langle u, v \rangle)) =$$



$$x + \bar{x} = 2\operatorname{Re}(x)$$

$$= 4\operatorname{Re}(\langle u, v \rangle)$$

tedy:

$$\|u+v\|^2 - \|u-v\|^2 = 4\operatorname{Re}(\langle u, v \rangle)$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2)$$

2. Imaginarní část sdělné podoby

$$\begin{aligned} \|u - iv\|^2 - \|u + iv\|^2 &= \|u\|^2 + \|v\|^2 + \\ &+ \langle u, -iv \rangle + \langle -iv, u \rangle - (\|u\|^2 + \|v\|^2 + \\ &+ \langle u, iv \rangle + \langle iv, u \rangle) = \\ &= -i \langle u, v \rangle + \overline{-i \langle u, v \rangle} - \end{aligned}$$

Použijeme lineárity a spárovaní

$$\begin{aligned} - (i \langle u, v \rangle + \overline{i \langle u, v \rangle}) &= \\ &= i (\langle \overline{u}, v \rangle - \langle u, v \rangle) = \\ &= i (\langle \overline{u}, v \rangle - \langle u, v \rangle) = \end{aligned}$$

Poznámka: $\overline{\overline{x}} = x = -2i \operatorname{Im}(x)$

$$= 2 \operatorname{Im}(\langle \overline{u}, v \rangle) + 2 \operatorname{Im}(\langle u, v \rangle) = 4 \operatorname{Im}(\langle u, v \rangle)$$

$$\|u - iv\|^2 - \|u + iv\|^2 = 4 \operatorname{Im}(\langle u, v \rangle)$$

Problema

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2)$$